SIMATS ENGINEERING

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Engineer to Excel

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Course Code: CSA1616 **Slot:** D

Course Name: Data warehouse and Data mining

Course Faculty: Dr. Mahesh Muthulakshmi

Experiment:22

AIM:- To show data set and perform the Apriori Algorithm and FP algorithm s

Procedure-

1.Open the notepad and select the new page.

2.enter the ARFF code for following question .

3. save the file as example -exp24.arff.

4. open the weka compiler.

5.select the explore option.

6. open the file .

7. select the associate option .

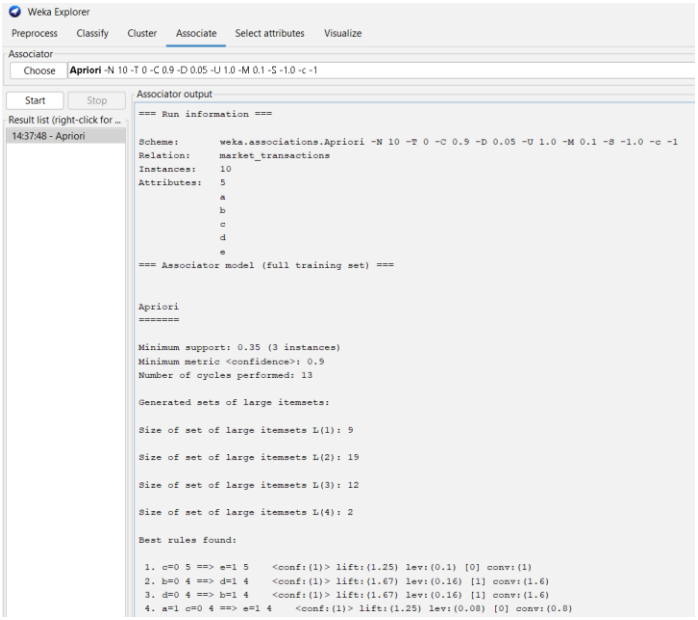
8. select apriori edit the min support and min mertic.

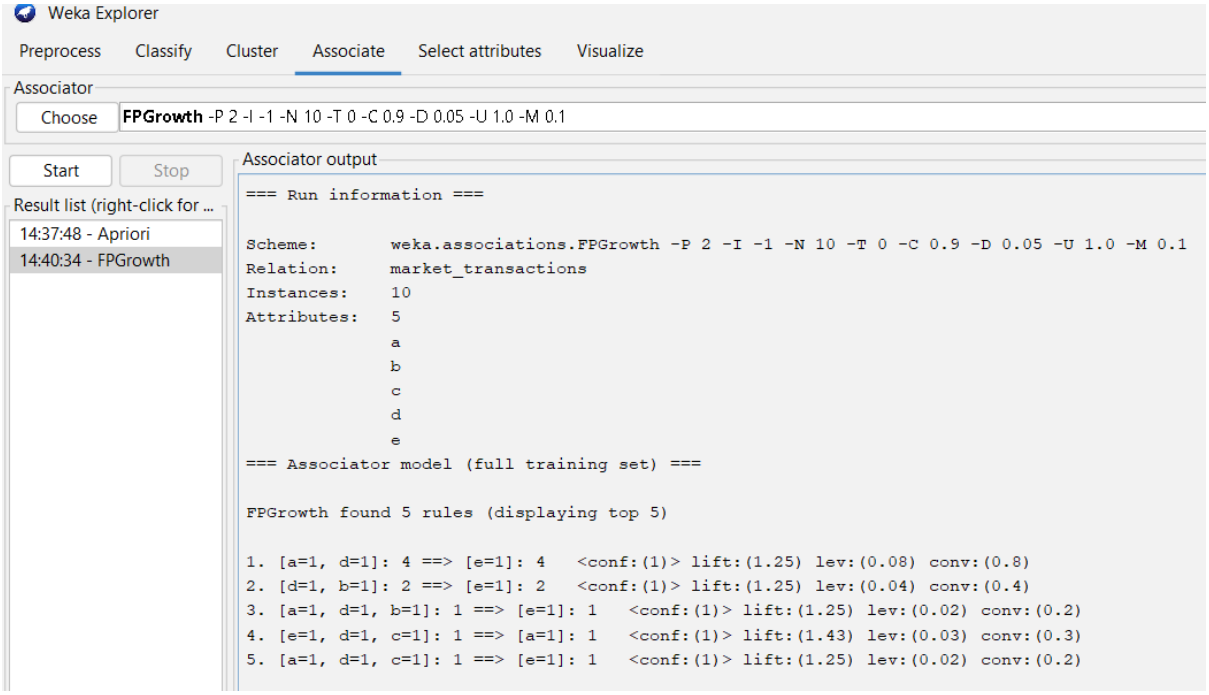
9. clik the start it displays output for apriori algorithm.

10.select FP-growth edit the min support

11. clik the start it displays output for FP-growth.

Output:





Experiment-23

Aim:- :- To show data set and perform the Apriori Algorithm and FP algorithm s

Procedure-

1.Open the notepad and select the new page.

2.enter the ARFF code for following question .

3. save the file as example -exp24.arff.

4. open the weka compiler.

5.select the explore option.

6. open the file .

7. select the associate option .

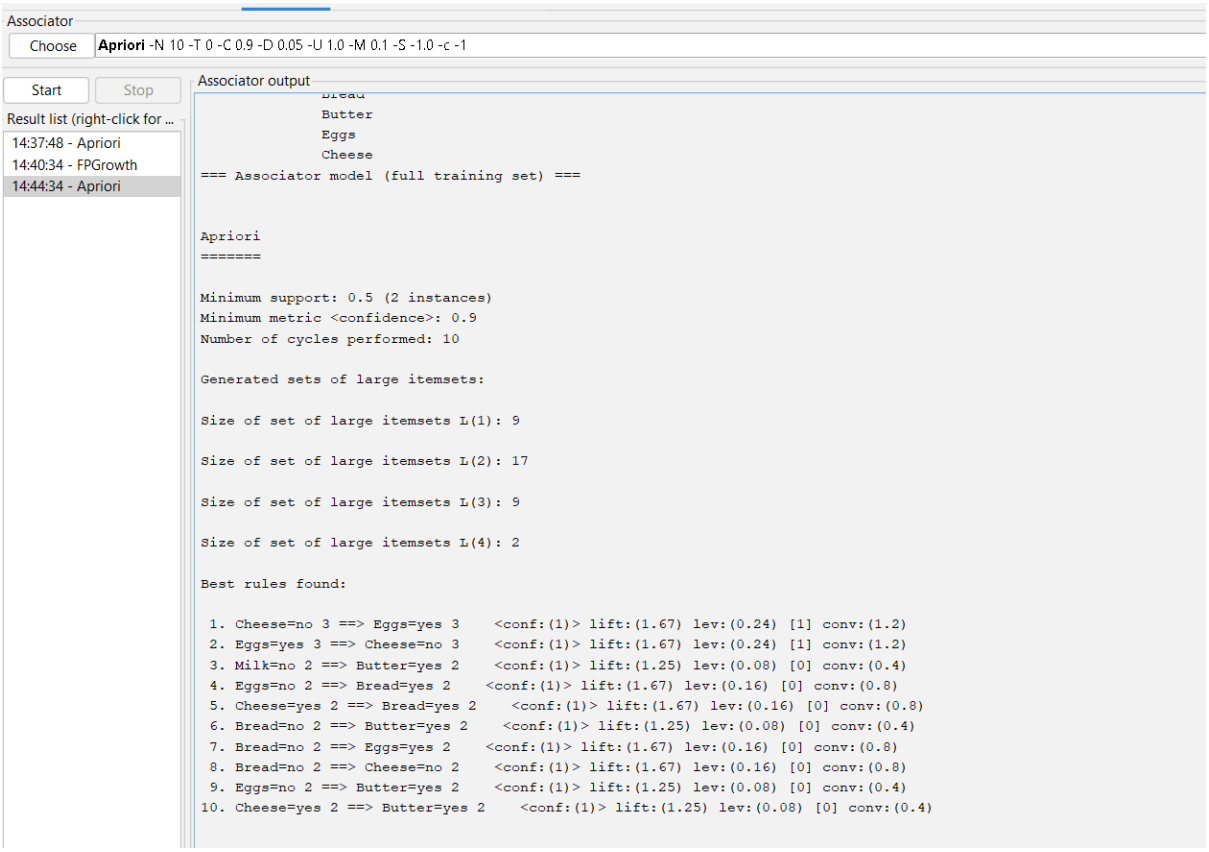
8. select apriori edit the min support and min mertic.

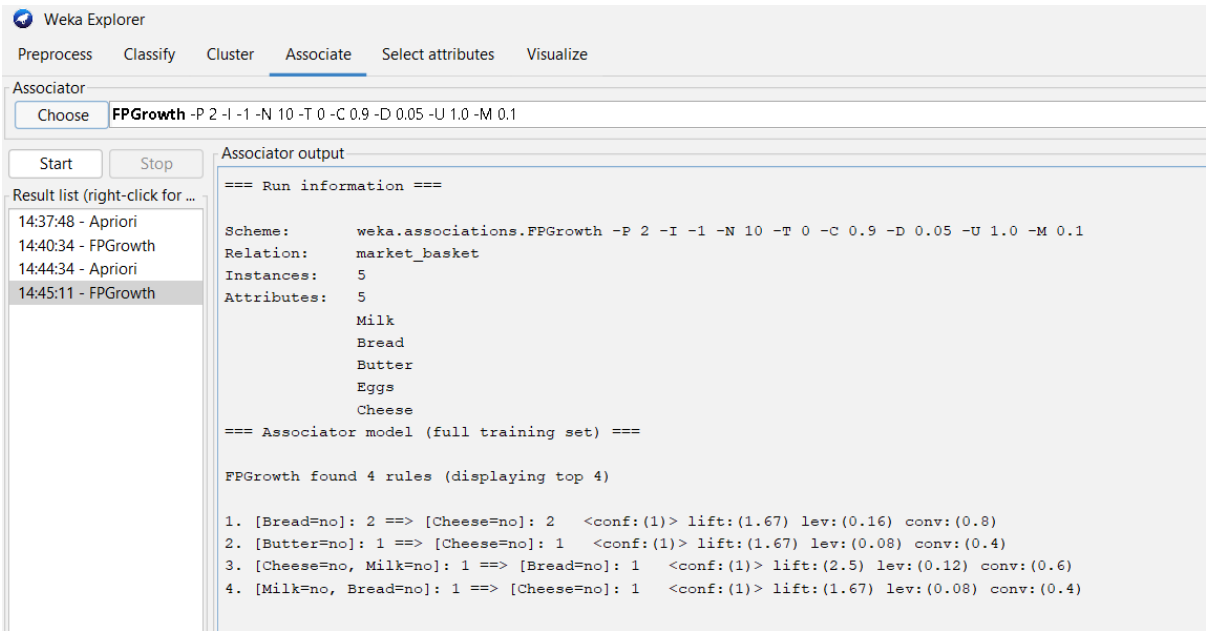
9. clik the start it displays output for apriori algorithm.

10.select FP-growth edit the min support

11. clik the start it displays output for FP-growth.

Output:


Associator
Choose
Apriori -N 10 -T 0 -C 0.9 -D 0.05 -U 1.0 -M 0.1 -S -1.0 -c -1
Start
Stop
Result list (right-click for ...
14:37:48 - Apriori
14:40:34 - FPGrowth
14:44:34 - Apriori
Associator output
Bread
Butter
Eggs
Cheese
=== Associator model (full training set) ===
Apriori
=======
Minimum support: 0.5 (2 instances)
Minimum metric <confidence>: 0.9
Number of cycles performed: 10
Generated sets of large itemsets:
Size of set of large itemsets L(1): 9
Size of set of large itemsets L(2): 17
Size of set of large itemsets L(3): 9
Size of set of large itemsets L(4): 2
Best rules found:
1. Cheese=no 3 ==> Eggs=yes 3 <conf:(1)> lift:(1.67) lev:(0.24) [1] conv:(1.2)
2. Eggs=yes 3 ==> Cheese=no 3 <conf:(1)> lift:(1.67) lev:(0.24) [1] conv:(1.2)
3. Milk=no 2 ==> Butter=yes 2 <conf:(1)> lift:(1.25) lev:(0.08) [0] conv:(0.4)
4. Eggs=no 2 ==> Bread=yes 2 <conf:(1)> lift:(1.67) lev:(0.16) [0] conv:(0.8)
5. Cheese=yes 2 ==> Bread=yes 2 <conf:(1)> lift:(1.67) lev:(0.16) [0] conv:(0.8)
6. Bread=no 2 ==> Butter=yes 2 <conf:(1)> lift:(1.25) lev:(0.08) [0] conv:(0.4)
7. Bread=no 2 ==> Eggs=yes 2 <conf:(1)> lift:(1.67) lev:(0.16) [0] conv:(0.8)
8. Bread=no 2 ==> Cheese=no 2 <conf:(1)> lift:(1.67) lev:(0.16) [0] conv:(0.8)
9. Eggs=no 2 ==> Butter=yes 2 <conf:(1)> lift:(1.25) lev:(0.08) [0] conv:(0.4)
10. Cheese=yes 2 ==> Butter=yes 2 <conf:(1)> lift:(1.25) lev:(0.08) [0] conv:(0.4)


Weka Explorer
Preprocess
Classify
Cluster
Associate
Select attributes
Visualize
Associator
Choose
FPGrowth -P 2 -I -1 -N 10 -T 0 -C 0.9 -D 0.05 -U 1.0 -M 0.1
Start
Stop
Result list (right-click for ...
14:37:48 - Apriori
14:40:34 - FPGrowth
14:44:34 - Apriori
14:45:11 - FPGrowth
Associator output
=== Run information ===
Scheme: weka.associations.FPGrowth -P 2 -I -1 -N 10 -T 0 -C 0.9 -D 0.05 -U 1.0 -M 0.1
Relation: market_basket
Instances: 5
Attributes: 5
Milk
Bread
Butter
Eggs
Cheese
=== Associator model (full training set) ===
FPGrowth found 4 rules (displaying top 4)
1. [Bread=no]: 2 ==> [Cheese=no]: 2 <conf:(1)> lift:(1.67) lev:(0.16) conv:(0.8)
2. [Butter=no]: 1 ==> [Cheese=no]: 1 <conf:(1)> lift:(1.67) lev:(0.08) conv:(0.4)
3. [Cheese=no, Milk=no]: 1 ==> [Bread=no]: 1 <conf:(1)> lift:(2.5) lev:(0.12) conv:(0.6)
4. [Milk=no, Bread=no]: 1 ==> [Cheese=no]: 1 <conf:(1)> lift:(1.67) lev:(0.08) conv:(0.4)

Experiment-24

Aim:- To show Bayes classification and descion tree.

Procedure-

1.Open the notepad and select the new page.

2.enter the ARFF code for following question .

3. save the file as example -exp24.arff.

4. open the weka compiler.

5.select the explore option.

6. open the file .

7. select the classify option .

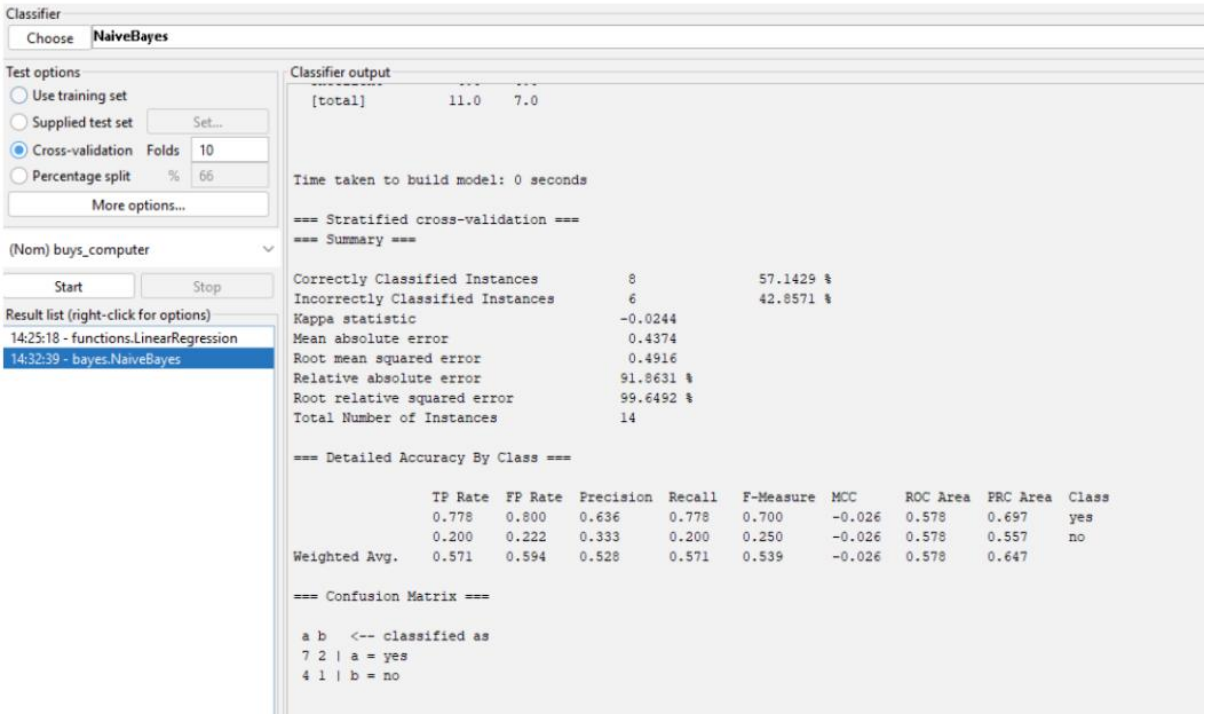
8. select Navi bayes.

9. clik the start it displays output for Navi bayes algorithm.

10.select J48

11. clik the start it displays output for J48.

Output:



Classifier output

```
J48 pruned tree
------------------

age = <=30
|   student = yes: yes (2.0)
|   student = no: no (3.0)
age = 31-40: yes (4.0)
age = >40
|   credit_rating = fair: yes (3.0)
|   credit_rating = excellent: no (2.0)

Number of Leaves  :     5

Size of the tree :      8


Time taken to build model: 0.02 seconds

=== Stratified cross-validation ===
=== Summary ===

Correctly Classified Instances           7               50      %
Incorrectly Classified Instances         7               50      %
Kappa statistic                         -0.0426
Mean absolute error                      0.4167
Root mean squared error                  0.5984
Relative absolute error                 87.5    %
Root relative squared error            121.2987 %
Total Number of Instances               14

=== Detailed Accuracy By Class ===
```

